Sample Paper 12

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

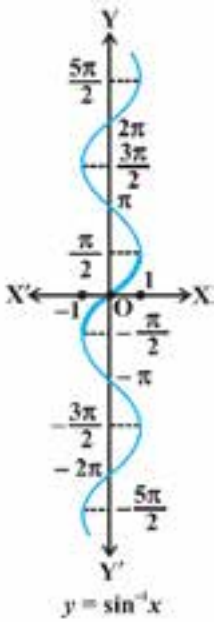
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. At which point does the graph of $y = \cos^{-1} x$ intersect the Y-axis?



(a) (0,0) (b) $(0, \pi/2)$

(c) $(0, \pi)$ (d) (0,1)

2. Integrating factor (IF) of the differential equation

$$\frac{dy}{dx} - y\cos x = \sin x \cos x$$

(a) $e^{-\sin x}$ (b) $e^{\sin x}$

(c) $e^{-\cos x}$ (d) $e^{\cos x}$

3. The solution of the differential equation $\frac{dy}{dx} = \frac{x}{y}$ is

(a) $x - y = k$ (b) $x^2 - y^2 = k$

(c) $x^3 - y^3 = k$ (d) $xy = k$

4. $\frac{d}{dx}(\sin x) =$

(a) $\cos x$ (b) $-\sin x$

(c) $-\cos x$ (d) $\tan x$

5. $\hat{i} \times (\hat{i} \times \hat{j}) + \hat{j} \times (\hat{j} \times \hat{k}) + \hat{k} \times (\hat{k} \times \hat{i}) =$

(a) $\hat{i} + \hat{j} + \hat{k}$ (b) 0

(c) 1 (d) $-(\hat{i} + \hat{j} + \hat{k})$

6. The real number which most exceeds its cube is

(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{2}$

(c) $\frac{1}{\sqrt{2}}$ (d) None of these

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7. If the direction cosines of two straight lines are l_1, m_1, n_1 and l_2, m_2, n_2 then the cosine of the angle θ between them or $\cos\theta$ is

(a) $(l_1+m_1+n_1)(l_2+m_2+n_2)$

(b) $\frac{l_1}{l_2}+\frac{m_1}{m_2}+\frac{n_1}{n_2}$

(c) $l_1l_2+m_1m_2+n_1n_2$

(d) $\frac{l_1+m_1+n_1}{l_2+m_2+n_2}$

8. If $\lambda \in R$ and $\Delta = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$ then $\lambda\Delta =$

(a) $\begin{vmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{vmatrix}$ (b) $\begin{vmatrix} \lambda a & b \\ c & d \end{vmatrix}$

(c) $\begin{vmatrix} \lambda a & b \\ \lambda c & d \end{vmatrix}$ (d) None of these

9. The direction ratios of a straight line are 1,3,5. Its direction cosines are

(a) $\frac{1}{\sqrt{35}},\frac{3}{\sqrt{35}},\frac{5}{\sqrt{35}}$ (b) $\frac{1}{9},\frac{1}{3},\frac{5}{9}$

(c) $\frac{3}{\sqrt{35}},\frac{5}{\sqrt{35}},\frac{1}{\sqrt{35}}$ (d) $\frac{5}{\sqrt{35}},\frac{3}{\sqrt{35}},\frac{1}{\sqrt{35}}$

10. $\int \frac{xe^x}{(x+1)^2}dx =$

(a) $\frac{e^x}{(x+1)^2}+c$ (b) $\frac{-e^x}{x+1}+c$

(c) $\frac{e^x}{x+1}+c$ (d) $\frac{-e^x}{(x+1)^2}+c$

11. Maximum value of $f(x)=\sin x+\cos x$ is

(a) 1 (b) 2

(c) $\frac{1}{\sqrt{2}}$ (d) $\sqrt{2}$

12. $f(x)=2x^3-15x^2+36x+4$ is

(a) increasing in $(-\infty, 2]$

(b) increasing in [2, 3]

(c) decreasing in $(3,\infty)$

(d) None of these

13. The optimal value of the objective function is attached at the point:

(a) given by intersection of in-equations with axes only.

(b) given by intersection of in-equations with x-axis only.

(c) given by corner points of the feasible region.

(d) none of the above.

14. If $A=\{1,2,3\}$, then how many equivalence relation can be defined on A containing (1, 2) :

(a) 2 (b) 3

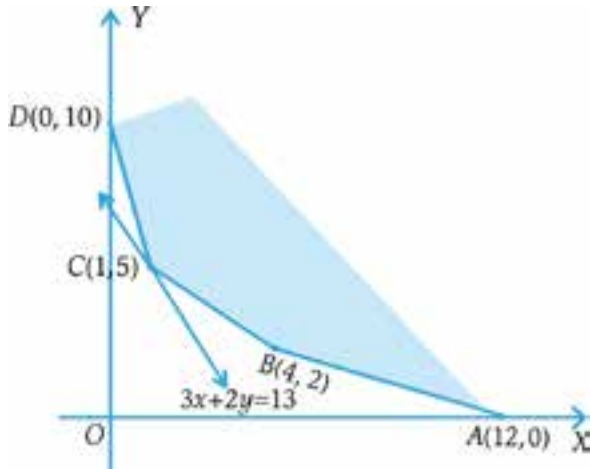
(c) 8 (d) 6

15. $\frac{d}{dx}[\log x] = ?$

(a) $\frac{1}{x}$ (c) $-\frac{1}{x^2}$

(c) 1 (d) $\frac{1}{x^2}$

16. Using the feasible region shown in Fig., determine the maximum value of the objective function $z=2x+3y$. The feasible region has corner points A(12,0), B(4,2), C(1,5), and D(0,10).

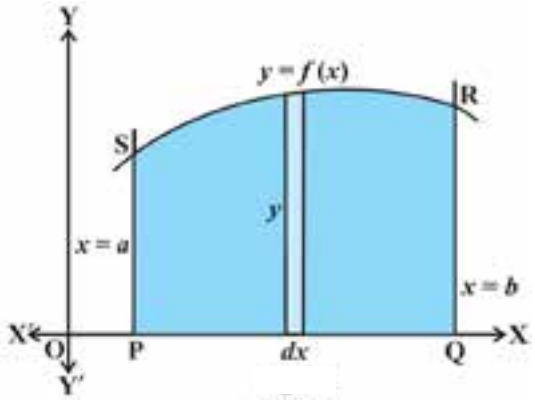


(a) 24 (b) 14

(c) 17 (d) 30

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17. In the given figure, the shaded region lies inside which curve?



(a) Circle with center O and radius a

(b) Ellipse with semi-major axis a and semi-minor axis b

(c) Parabola opening along the X-axis

(d) Hyperbola with foci on the X-axis

18. $\int_2^1 \frac{dx}{x} = ?$

(a) $\log\frac{2}{3}$ (b) $\log\frac{3}{2}$

(c) $\log\frac{1}{2}$ (d) $\log\frac{x}{2}$

19. **Assertion :** If A and B are two independent events $P(A \cup B) = 1 - P(A')P(B')$

Reason : $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ and $P(A \cap B) = P(A)P(B)$

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

20. **Assertion :** $f(x)$ is defined as $f(x) = \begin{cases} x^3 - 3, x \le 2 \\ x^2 + 1, x > 2 \end{cases}$ is continuous at $x = 2$.

Reason : $f(2) = \lim_{x \to 2} f(x)$.

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Find $\frac{dy}{dx}$ if $y = \cos\sqrt{\sin x}$

OR

Find $\frac{dy}{dx}$, when $x = y\log(xy)$

22. Show that the line joining the points (4, 7, 8), (2, 3, 4) is parallel to the line joining the points (2, 4, 10), (–2, –4, 2).

Two lines are said to be parallel if direction ratio of points forming the line are proportional to each other.

Direction Ratio of line joining $P_1(4,7,8)$ and $P_2(2,3,4)$ is

$$DR_1 = (4-2, 7-3, 8-4)$$
$$= (2,4,4)$$

Direction Ratio of line joining $P_3(2,4,10)$ and $P_4(-2,-4,2)$ is

$$DR_2 = (2+2, 4+4, 10-2)$$
$$= (4,8,8)$$

Since direction ratio are proportional to each other with ratio 2, hence lines formed by (P_1, P_2) and (P_3, P_4) are parallel to each other.

23. Prove that $-|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - |\vec{a}\,\vec{b}|^2$

24. If A and B are two independent events then prove that : $P(A \cup B) = 1 - P(A') \cdot P(B')$

25. Solve : $\frac{dy}{dx} + 2y\tan x = \sin x$.

OR

Show that the function $1 + 8y^2 \tan x = ay^2$ is a solution of differential equation

$$\cos^2 x \frac{dy}{dx} = 4y^3$$

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

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26. Find the value of p, if
$(2\hat{i}+6\hat{j}+27\hat{k})\times(\hat{i}+3\hat{j}+p\hat{k})=0$

27. Find the value of x, such that

$$\begin{bmatrix}1&0&2\\0&2&1\\2&0&3\end{bmatrix}\begin{bmatrix}x\\4\\1\end{bmatrix}\begin{bmatrix}x&-5&-1\end{bmatrix}=0$$

OR

If $A=\begin{bmatrix}2&5\\3&1\end{bmatrix}$ and $B=\begin{bmatrix}1&5\\6&2\end{bmatrix}$ then find $(A+B)$ and $(A-B)$.

28. The radius of a circle is increasing uniformly at the rate of 3 cm/sec. Find the rate at which the area of the circle is increasing when the radius is 10 cm.

29. If $f(x)=x\sin\frac{1}{x}$, when $x\neq 0$; and, $f(x)=0$, when $x=0$, then test the continuity of $f(x)$ at $x=0$.

30. Find the value of $\cot^{-1}\left(\tan\frac{\pi}{7}\right)$?

OR

Prove that $4(\cot^{-1}3+\operatorname{cosec}^{-1}\sqrt{5})=\pi$.

31. In the set Q of all rational numbers, a binary operation $o: Q\times Q\to Q$ is defined by $o(x,y)=x\,o\,y=x+y-xy$ then show that o is commutative.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Evaluate $\int\frac{-7x+2}{\sqrt{16x^2-9}}dx$

OR

Prove that $\int_0^{\pi/2}\log\sin x\,dx=\int_0^{\pi/2}\log\cos x\,dx=-\frac{\pi}{2}\log 2$

33. If $x=\cos t+\log\tan\left(\frac{t}{2}\right)$, $y=\sin t$, then find the values of $\frac{d^2y}{dt^2}$ and $\frac{d^2y}{dx^2}$ at $t=\frac{\pi}{4}$.

OR

Find the values of a and b, if the function f defined by $f(x)=\begin{cases}x^2+3x+a, & x\leq 1\\ bx+2, & x>1\end{cases}$ is differentiable at $x=1$.

34. Solve the following L.P. Problem graphically:

Maximise $Z=x+y$

Subject to $x-y\leq -1$

$-x+y\leq 0$

$x,\ y\geq 0$

35. Solve : $\frac{dy}{dx}-\frac{2y}{x}=y^4$

OR

Solve $y^2\,dx+(x^2+xy)\,dy=0$

SECTION - E

Case study based questions are compulsory.

36. Rice is a nutritional staple food which provides instant energy as its most important component is carbohydrate (starch). On the other hand, rice is poor in nitrogenous substances with average composition of these substances being only 8 per cent and fat content or lipids only negligible, i.e., 1per cent and due to this reason it is considered as a complete food for eating. Rice flour is rich in starch and is used for making various food materials.



Two farmers Ramkishan and Gurcharan Singh cultivate only three varieties of rice namely Basmati, Permal and Naura. The sale (in ₹) of these varieties of rice by both the farmers in the month of September and October are given by the following matrices A and B.

September Sales (in ₹)

$$A=\begin{array}{c}\begin{matrix}\text{Basmati}&\text{Permal}&\text{Naura}\end{matrix}\\\begin{bmatrix}10000&20000&30000\\50000&30000&10000\end{bmatrix}\begin{matrix}\text{Ramkishan}\\\text{Gurcharan Singh}\end{matrix}\end{array}$$

October Sales (in ₹)

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$$B = \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ \end{matrix}$$
$$\begin{bmatrix} 5000 & 10000 & 6000 \\ 20000 & 10000 & 10000 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

(i) Find the combined sales in September and October for each farmer in each variety.

(ii) Find the decrease in sales from September to October.

(iii) If both farmers receive 2% profit on gross sales, compute the profit for each farmer and for each variety sold in October.

37. Western music is a form of country music composed by and about the people who settled and worked throughout the Western United States and Western Canada. Western music celebrates the lifestyle of the cowboy on the open ranges, Rocky Mountains, and prairies of Western North America.



Western music is organised every year in the stadium that can hold 36000 spectators. With ticket price of ` 10, the average attendance has been 24000. Some financial expert estimated that price of a ticket should be determined by the function $p(x) = 15 - \frac{x}{3000}$, where x is the number of ticket sold.

Bases on the above information, answer of the following questions.

(i) Find the expression for total revenue R as a function of x.

(ii) Find the value of x for which revenue is maximum.

(iii) When the revenue is maximum, what will be the price of the ticket?

OR

(iv) How many spectators should be present to maximum the revenue?

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38. If two vectors are represented by the two sides of a triangle taken in order, then their sum is represented by the third side of the triangle taken in opposite order and this is known as triangle law of vector addition.

Based on the above information, answer the following questions.

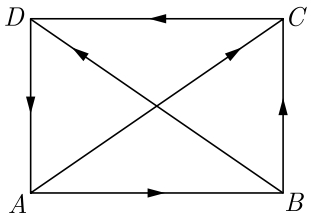
(i) If $\hat{p}$, $\hat{q}$, $\hat{r}$ are the vectors represented by the side of a triangle taken in order, then find $\vec{q} + \vec{r}$.

(ii) If $ABCD$ is a parallelogram and AC and BD are its diagonals, then find $\overrightarrow{AC} + \overrightarrow{BD}$.

(iii) If $ABCD$ is a parallelogram, where $\overrightarrow{AB} = 2\vec{a}$ and $\overrightarrow{BC} = 2\vec{b}$, then find $\overrightarrow{AC} - \overrightarrow{BD}$.

OR

(iv) If ABCD is a quadrilateral, whose diagonals are $\overrightarrow{AC}$ and $\overrightarrow{BD}$, then find $\overrightarrow{BA} + \overrightarrow{CD}$.



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